



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES
ITEP 414



System Administration and Maintenance

Week 2 Portfolio Project

**Enterprise Infrastructure Planning
for a
Startup Company**

MATIENZO, JOHN LESTER D.

BSIT 3C

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Part 1. Company Profile

ABC Startup Solutions

Company Name

Nature of Business

ABC Startup Solutions is a newly established software development company that provides customized software applications, web-based systems, database solutions, and technical support services for small and medium-sized businesses.

Company Vision

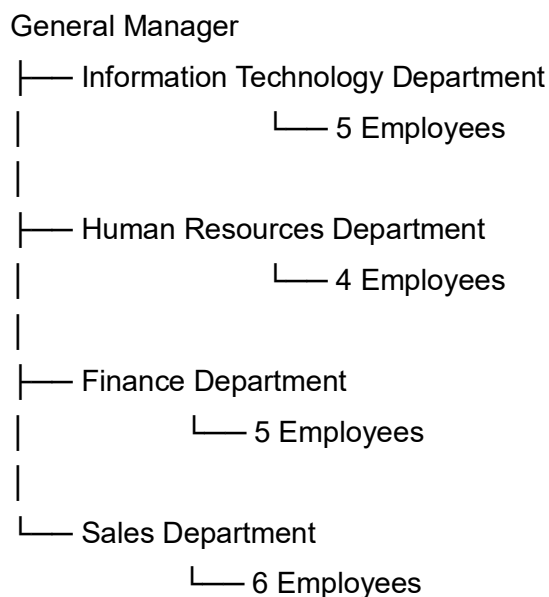
To become a trusted software development company that provides reliable, affordable, and innovative IT solutions for businesses.

Company Mission

ABC Startup Solutions aims to provide quality software and technical services by using reliable technologies, maintaining secure IT infrastructure, and providing responsive support to clients.

Office Location: Bagong Pook Business Center, Pila, Laguna, Philippines.

Organizational Structure





Employee Distribution

Department	Number of Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

Part 2. Hardware Inventory

This part identifies the hardware needed to support ABC Startup Solutions. The equipment is selected based on the company's 20 employees and the needs of the IT, HR, Finance, and Sales departments. Each item and quantity is planned to support daily operations, networking, storage, backup, and power protection.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	20	All Departments	Used by employees for daily work, communication, documentation, software development, finance, HR, and sales activities.
HW-002	Laptop	5	IT and Sales	Used for meetings, presentations, remote work, troubleshooting, and mobility.
HW-003	Server	1	IT	Hosts company services, applications, files, and internal systems.
HW-004	Router	1	IT	Connects the company network to the internet and manages network traffic.
HW-005	Managed Switch	1	IT	Connects computers, server, printer, access point, and other network devices.



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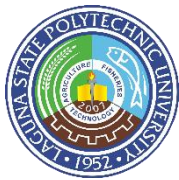


HW-006	Network Printer	2	Finance and HR	Provides shared printing for documents, reports, and company records.
HW-007	UPS	3	IT	Provides power protection for the server, network equipment, and important workstations.
HW-008	Wireless Access Point	2	All Departments	Provides wireless network access throughout the office.
HW-009	NAS Storage	1	IT	Provides centralized file storage and backup storage.
HW-010	External Backup Drive	2	IT	Stores additional offline backups of important company data.
HW-011	Monitor	20	IT and Other Departments	Provides display devices for desktop and laptop workstations.

Part 3. Software Inventory

This part identifies the software needed to support ABC Startup Solutions. The software is selected based on the company's daily operations, software development, system administration, security, communication, and file management needs. Each software is listed with its version, license, and purpose.

Software	Version	License	Purpose
Windows 11 Pro	Windows 11 Pro 24H2	Commercial	Main operating system for employee computers.
Ubuntu Server	Ubuntu Server 24.04 LTS	Open Source	Used as the company's server operating system.
Microsoft Office	Microsoft 365 Apps	Subscription	Used for documents, spreadsheets, presentations, and office work.



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Visual Studio Code	Latest Stable Version	Free	Used by IT employees for programming and software development.
Git	Latest Stable Version	Open Source	Used for source code version control.
GitHub Desktop	Latest Stable Version	Free	Provides a graphical interface for Git and GitHub repositories.
VirtualBox	Version 7.x	Open Source	Used for creating virtual machines for testing and learning.
Google Chrome	Latest Stable Version	Free	Used for web browsing and accessing web-based business systems.
Microsoft Defender	Included with Windows 11 Pro	Included	Provides antivirus and security protection for Windows computers.
AnyDesk	Latest Stable Version	Commercial/Free options	Used by IT staff for remote technical support.
7-Zip	Latest Stable Version	Open Source	Used to compress and extract files.



Part 4. Network Inventory

This part identifies the networking equipment needed to connect and protect ABC Startup Solutions' office network. The inventory includes the ISP modem, router, firewall, switch, access points, patch panel, CAT6 cables, and RJ45 connectors required for the company's network infrastructure.

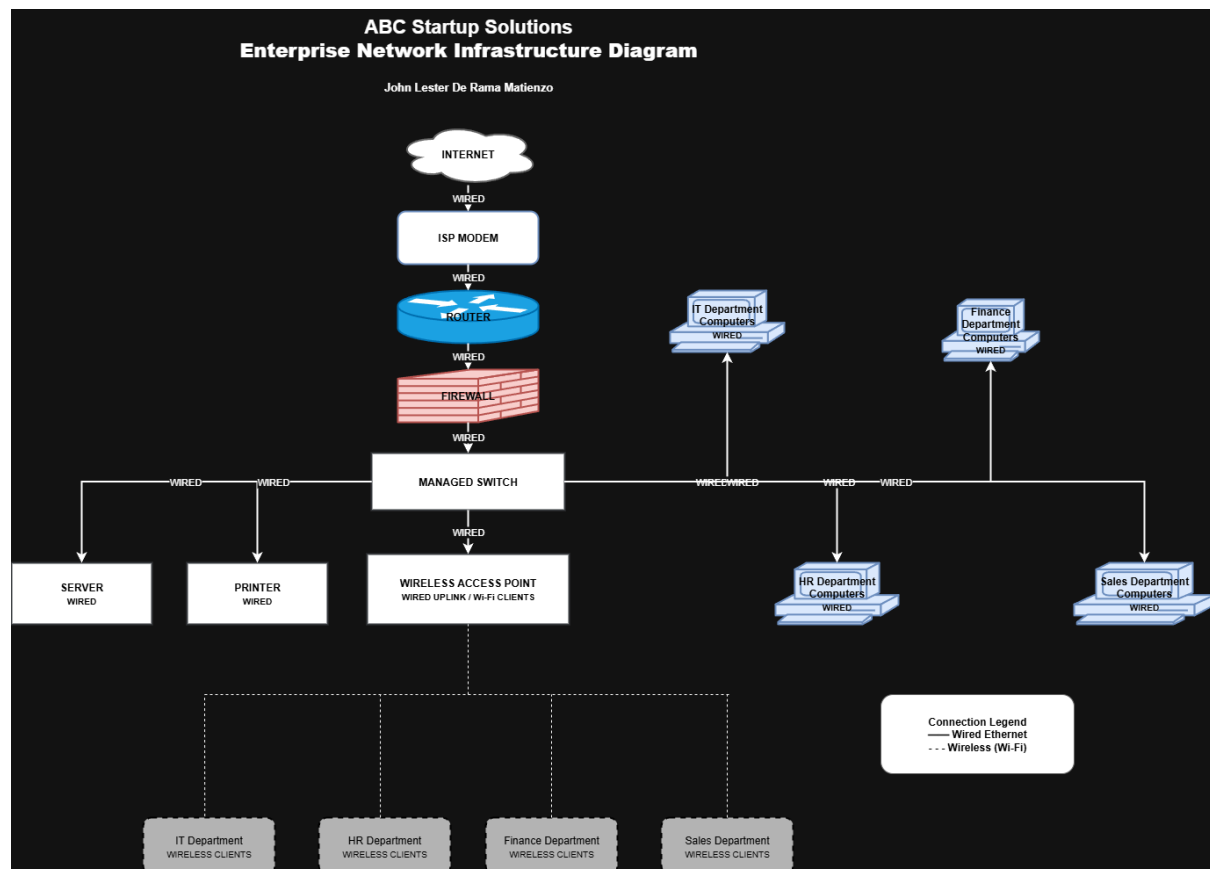
Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Receives the internet connection from the Internet Service Provider.
NET-002	Router	1	Routes traffic between the company network and the internet.
NET-003	Firewall	1	Protects the internal network from unauthorized network traffic.
NET-004	Managed Switch	1	Connects wired computers, server, printer, and other network devices.
NET-005	Wireless Access Point	2	Provides wireless connectivity for laptops and other devices.
NET-006	Patch Panel	1	Organizes and manages Ethernet connections in the network cabinet.
NET-007	CAT6 Cable	500 meters	Provides wired network connections between devices and network infrastructure.
NET-008	RJ45 Connectors	50 pieces	Used to terminate CAT6 Ethernet cables.

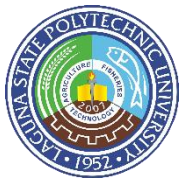


Part 5. Network Diagram

This part presents the enterprise network infrastructure of ABC Startup Solutions, showing how devices and departments are connected from the Internet through the ISP modem, router, firewall, managed switch, and wireless access point.

The diagram illustrates the logical flow of network communication within the company, including the server, printer, wireless access point, and computers for the IT, HR, Finance, and Sales departments, with both wired and wireless connections shown.





Part 6. System Administration Roles

1. Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-level technical support to employees. The technician troubleshoots hardware, software, account, printer, and basic network problems. The technician also documents reported problems and their solutions.

Skills

- Computer hardware and software troubleshooting
- Operating system knowledge
- Basic networking
- Communication skills
- Problem-solving
- Customer service

Common Tools

- Windows troubleshooting tools
- Remote desktop software
- AnyDesk
- Ticketing systems
- Command Prompt
- PowerShell

Certifications

- CompTIA A+
- ITIL Foundation
- CompTIA Network+



2. Network Administrator

Responsibilities

A Network Administrator manages and maintains the company's network infrastructure. The administrator configures routers, switches, firewalls, access points, and network security settings. The administrator also monitors network performance and resolves connectivity problems.

Skills

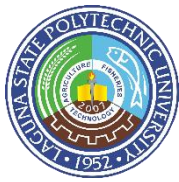
- TCP/IP networking
- Routing and switching
- Network security
- Troubleshooting
- Firewall configuration
- Network monitoring

Common Tools

- Cisco Packet Tracer
- Wireshark
- Ping
- Traceroute
- Network monitoring tools
- Router and switch management interfaces

Certifications

- CompTIA Network+
- Cisco Certified Network Associate
- CompTIA Security+



3. Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux servers and services. The administrator installs and configures Linux systems, manages users and permissions, monitors server performance, performs updates, and maintains server security.

Skills

- Linux command line
- User and permission management
- Shell scripting
- Server configuration
- File system management
- System troubleshooting

Common Tools

- Ubuntu Server
- Bash
- SSH
- systemctl
- cron
- Linux terminal

Certifications

- Linux+
- LPIC-1
- Red Hat Certified System Administrator



4. Cloud Administrator

Responsibilities

A Cloud Administrator manages cloud-based infrastructure and services. The administrator creates and manages virtual machines, storage, networking, user permissions, backups, and cloud security.

Skills

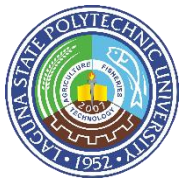
- Cloud computing
- Virtualization
- Networking
- Cloud security
- Storage management
- Identity and access management

Common Tools

- Microsoft Azure
- Amazon Web Services
- Google Cloud
- Azure Portal
- AWS Management Console
- Cloud monitoring tools

Certifications

- Microsoft Certified: Azure Administrator Associate
- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect



How the Four Professionals Work Together

The four professionals work together to maintain the ABC Startup Solutions 's IT environment. The Helpdesk Technician handles employee technical problems and performs initial troubleshooting. When an issue involves network connectivity, routers, switches, firewalls, or wireless connections, the Network Administrator handles the problem. The Linux System Administrator manages the company's Linux server and server-related services. The Cloud Administrator manages cloud-based resources and services. By working together, these professionals help maintain reliable, secure, and available IT services for the organization.

The module specifically requires responsibilities, skills, common tools, certifications, and a final explanation of how the four roles work together.

Part 7. Infrastructure Recommendations

- **Internet Provider**

I recommend a business fiber internet connection with at least 300 Mbps download and upload capacity. A business plan is suitable for the startup because all departments need internet access for communication, cloud applications, software development, online meetings, and other business activities. The connection should also include a service agreement with the provider for technical support and service restoration.

- **Server Specifications**

I recommend the following initial server configuration:

- Processor: Intel Xeon or AMD EPYC class processor
- RAM: 32 GB
- Storage: 2 × 1 TB SSD in RAID 1
- Operating System: Ubuntu Server LTS
- Network: Gigabit Ethernet
- Backup: Connected to NAS storage



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This configuration provides sufficient resources for a small startup's initial server requirements while allowing future upgrades.

- **Backup Strategy**

I recommend using a 3-2-1 backup strategy. Important company files should have three copies, stored on two different types of storage, with one copy kept separately from the main system.

The company should perform daily backups of important files and server data. Weekly backups should also be retained for longer-term recovery. The NAS should hold regular backups, while external backup drives should provide an additional offline copy.

- **Security Recommendations**

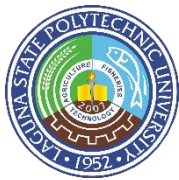
The company should deploy a firewall between the internal network and the internet. User accounts should have individual credentials and appropriate access permissions. Operating systems and applications should receive regular security updates. Important systems should use multi-factor authentication where supported. Employees should also receive basic security awareness training.

- **Antivirus**

Microsoft Defender should be enabled on Windows 11 Pro computers. Security definitions and operating system updates should remain current. The IT department should monitor security alerts and respond to detected threats.

- **Password Policy**

The company should require employees to use strong and unique passwords. Passwords should not be shared with other employees. Employees should avoid using personal information in passwords. Multi-factor authentication should be enabled for important accounts and services. User



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accounts should also receive appropriate access permissions based on job responsibilities.

- **Expansion Plan**

The company should prepare for growth from 20 employees to at least 40 employees. The network should have additional switch capacity and available network ports. The server and NAS should support additional storage and memory. Additional wireless access points should be added if office coverage becomes insufficient. The internet plan should also be upgraded as the number of employees and online services increases.

The module requires recommendations for the internet provider, server specifications, backup strategy, security, antivirus, password policy, and expansion plan. It also requires logical reasons for the recommendations.

Part 8. Personal Reflection

Doing this Week 2 Portfolio Project, it taught me that planning first on how the infrastructure should be design is very important before fully implementing its IT environment. It also helped me on how to properly identify the hardware, software, and networking requirements of the ABC Startup Solutions based on its number of employees and its business operations. I also learned and gained a better understanding on how different system administration roles work together to maintain, support, and ensure the efficiency of a company/organization's IT infrastructure.

For me, the most challenging task was the part 4, 5, and 7. **Part 4** was challenging because although I already had some knowledge about networking equipment, I was not completely familiar with all the items included in the inventory, so I had to review their purposes and understand how they are used together in an enterprise network. **Part 5** was also challenging because I had some basic knowledge of network devices, but I needed to apply that knowledge to create a complete and logical network diagram. I had to determine the proper connection between the Internet, ISP modem, router, firewall, switch, server, printer, wireless access point, and the different departments. **Part 7** was the most unfamiliar part for me because I was



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new to creating infrastructure recommendations. I needed to learn how to properly recommend an Internet provider, server specifications, backup strategy, security measures, antivirus, password policies, and an expansion plan. This part required me to think beyond simply identifying equipment and understand how different decisions can affect the company's security, reliability, and future growth.

I also learned why proper planning is important when building an IT infrastructure. Planning allows an organization to identify its requirements before purchasing equipment and deploying systems. It can help prevent unnecessary expenses, reduce configuration problems, improve security, and provide a clear direction for implementing and maintaining the infrastructure.

Overall, this project helped me realize that being a System Administrator involves more than just troubleshooting computers and fixing technical problems. It also requires proper planning, understanding the organization's needs, selecting suitable hardware and software, organizing and documenting the infrastructure, and considering security and future expansion. Through this project, I gained new knowledge and a better understanding of how an IT infrastructure is planned and maintained. The experience and skills I gained will help me become more confident and prepared for a future career in IT support or system administration.

References

<https://paios-classroom.com/assignment-attachments/13/file>

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